Problem 5

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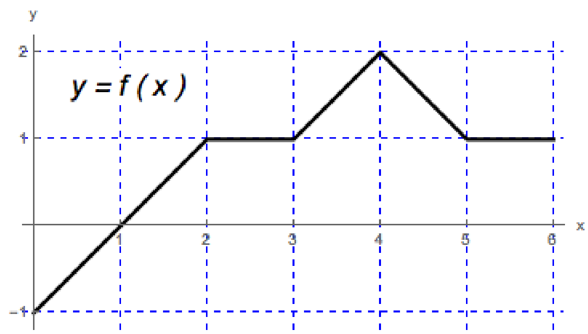
Problem 1 *The graph of a function f defined on the interval $[0, 6]$ is given in the figure.*



(a) *Compute the net area of the region between the graph of f and the x-axis, on the interval $[0, 6]$. [3]*

(b) *Draw a rectangle with base on the x-axis, $0 \leq x \leq 6$, whose area is equal to the net area in part (a).*

(c) *In the figure, mark a point c in $(0, 6)$ such that $f(c)$ is the height of the rectangle from part (b).*

(d) *Using the figure and parts (a-c), what is the relationship between the rectangle from part (b), the net area from part (a), and the average value of f on $[0, 6]$? [3]*